

Cost-Effectiveness of Intismeran Autogene Plus Pembrolizumab Versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV Melanoma: An Updated Early Economic Evaluation

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Abstract

Background: On August 19, 2026, Merck and Moderna announced positive topline results from the Phase 3 INTerpath-001 trial, establishing intismeran autogene (mRNA-4157/V940) plus pembrolizumab as the first mRNA-based individualized neoantigen therapy (INT) to succeed in a late-stage trial. Intismeran plus pembrolizumab significantly improved recurrence-free survival (RFS) and distant metastasis-free survival (DMFS) versus pembrolizumab alone in patients with completely resected stage IIIB–IV melanoma. However, the economic value of this novel combination remains unknown.

Objective: To estimate the cost-effectiveness of intismeran plus pembrolizumab versus pembrolizumab alone as adjuvant therapy for resected stage IIIB–IV melanoma from a US payer perspective.

Methods: A four-state partitioned survival model (recurrence-free → locoregional recurrence → distant metastasis → death) was developed with a lifetime horizon (40 years) and 3% annual discounting. Clinical efficacy data were derived from the 5-year follow-up of the Phase 2b KEYNOTE-942 trial. Parametric survival functions (log-normal) were calibrated to RFS, DMFS, and OS survival probabilities at 18, 24, 36, 48, and 60 months, with the ratio-constrained structure $RFS \leq DMFS \leq OS$ ensuring internal consistency. Health state utilities were derived from published melanoma EQ-5D literature (RF 0.83, LR 0.64, DM 0.55). Sensitivity analyses included probabilistic sensitivity analysis (1,000 iterations), deterministic sensitivity analysis, and scenario analyses.

Results: In the base case, with general population mortality constraint applied from 60 months, intismeran plus pembrolizumab yielded 11.10 QALYs at a cost of \$860,118, compared with 7.34 QALYs at \$713,157 for pembrolizumab alone. The incremental cost-effectiveness ratio was \$39,105/QALY gained. At a willingness-to-pay threshold of \$150,000/QALY, the probability of cost-effectiveness was 98.2% (93.7% at \$100,000/QALY). Results were robust across a wide range of sensitivity analyses.

Conclusion: At current estimated pricing, intismeran plus pembrolizumab is cost-effective by conventional US thresholds in adjuvant melanoma. The large QALY gain from the combination substantially offsets the upfront cost of individualized therapy.

Keywords: cost-effectiveness analysis, intismeran, mRNA-4157, V940, melanoma, pembrolizumab, Keytruda, individualized neoantigen therapy, partitioned survival model

1 Introduction

1.1 A New Reimbursement Problem

On August 19, 2026, Merck and Moderna announced positive topline results from the Phase 3 INTERpath-001 trial, establishing intismeran autogene (mRNA-4157/V940) plus pembrolizumab as the first mRNA-based individualized neoantigen therapy (INT) to succeed in a late-stage trial Merck & Moderna [2026]. The press release confirmed statistically significant improvements in RFS and DMFS, though numerical hazard ratios have not yet been disclosed. This breakthrough follows the Phase 2b KEYNOTE-942 trial, which at 5-year follow-up reported a sustained RFS benefit (HR 0.51, 95% CI 0.294–0.887), DMFS benefit (HR 0.411, 95% CI 0.200–0.843), and 5-year OS rates of 92.2% (combo) versus 71.3% (pembrolizumab alone) Khattak et al. [2026].

Despite these clinical advances, the economic implications of INTs remain poorly understood. These therapies combine the high costs of immune checkpoint inhibitors with the additional expense of individualized manufacturing, creating a reimbursement challenge unlike any in oncology.

1.2 Why INTs Differ from Conventional Oncology Pricing

Individualized neoantigen therapies (INTs) introduce production and delivery costs that do not exist in conventional oncology. Unlike fixed-composition drugs, each INT treatment is custom-manufactured for a single patient based on tumor sequencing data, requiring mRNA synthesis, lipid nanoparticle formulation, and quality assurance for each batch Ibragimova et al. [2025], Li et al. [2026]. These costs scale with each new patient rather than benefiting from traditional manufacturing economies of scale.

Jefferies analysts have estimated the cost of such vaccines at \$100,000–\$300,000 per patient Jefferies Equity Research [2026]. By comparison, cancer immunotherapies such as Keytruda® (pembrolizumab) have annual costs of approximately \$220,896 in the adjuvant setting Keytruda.com [2026]. The combination of a high-cost immunotherapy with an individualized vaccine creates a compound affordability challenge for payers, particularly in the adjuvant

setting where patients are recurrence-free at the time of treatment.

1.3 Existing Economic Evidence

A prior exploratory cost-effectiveness analysis of V940 plus pembrolizumab in adjuvant melanoma was presented at ISPOR 2024 Mclean and van Hest [2024]. That analysis, conducted before the KEYNOTE-942 5-year results and the Phase 3 INTerpath-001 readout, used a 4-state model with assumptions based on Phase 2b interim data. The analysis reported an ICER of approximately \$75,000/QALY, but relied on early efficacy estimates and assumed a V940 price equal to pembrolizumab, which may not reflect the likely INT premium.

1.4 Clinical Basis and Research Objectives

The clinical foundation for this analysis comes from the 5-year follow-up of the Phase 2b KEYNOTE-942 trial, which provides the most mature efficacy data available for intismeran Khattak et al. [2026]. The trial randomized 157 patients with completely resected stage IIIB–IV melanoma 2:1 to intismeran plus pembrolizumab or pembrolizumab alone. The primary endpoint was RFS; secondary endpoints included DMFS, OS, and safety. At the 5-year analysis, the combination showed a sustained RFS benefit with a hazard ratio of 0.51 (95% CI 0.294–0.887), a DMFS HR of 0.411 (95% CI 0.200–0.843), and OS data still maturing with 5-year OS rates of 92.2% (combo) versus 71.3% (pembrolizumab alone).

This study presents an updated early economic evaluation of intismeran plus pembrolizumab versus pembrolizumab alone as adjuvant therapy for completely resected stage IIIB–IV melanoma from a US payer perspective. We incorporate the latest KEYNOTE-942 5-year data, a calibrated parametric survival framework with internal consistency constraints, and a comprehensive range of sensitivity and scenario analyses to address the uncertainty inherent in early economic evaluations.

2 Methods

2.1 Model Structure

A partitioned survival model (PSM) was developed in Python with four mutually exclusive health states: recurrence-free (RF), locoregional recurrence (LR), distant metastasis (DM), and death, following the structure of prior melanoma cost-effectiveness models and standard PSM methodology Woods et al. [2017], Zhang et al. [2023], Bensimon et al. [2019]. Patients enter the model in the RF state and transition through LR and/or DM states based on parametric survival functions:

$$\text{RF}(t) = \text{RFS}(t)$$

$$\text{LR}(t) = \text{DMFS}(t) - \text{RFS}(t)$$

$$\text{DM}(t) = \text{OS}(t) - \text{DMFS}(t)$$

$$\text{Dead}(t) = 1 - \text{OS}(t)$$

This construction ensures $\text{RFS} \leq \text{DMFS} \leq \text{OS}$ at all times and maintains internal consistency without forcing a specific parametric form on the transition hazards.

2.2 Clinical Data

Survival probabilities at 18, 24, 36, 48, and 60 months were extracted from the published KEYNOTE-942 Kaplan–Meier curves Khattak et al. [2026]. Parametric survival models were fitted to these data points using weighted nonlinear least squares, following standard methodological guidance for survival extrapolation in health economic evaluation Latimer [2011, 2013], Ishak et al. [2013]. Model selection was performed by comparing five candidate distributions (exponential, Weibull, log-normal, log-logistic, generalized gamma) for each of the three underlying survival components (OS, $r_{\text{DM}} = \text{DMFS}/\text{OS}$, $r_{\text{LR}} = \text{RFS}/\text{DMFS}$) using Akaike Information Criterion (AIC), as recommended by NICE DSU TSD 14 Latimer [2011]. The log-normal distribution was selected as the base case because it achieved the best AIC for the OS curve of the combination arm (the most influential curve driving lifetime costs and

QALYs), was within 0.5 AIC of the best for all other curves, and preserved structural consistency across all 4,096 candidate distribution combinations (Supplementary Table S1). The fitted model was calibrated to ensure the constraint $RFS \leq DMFS \leq OS$ was satisfied at all time points. To address the external validity of long-term survival extrapolation, a general population mortality hazard floor was applied from 60 months onward, using the age- and sex-matched US 2019 life table (pooled 64% male/36% female, baseline age 61 years) Arias and Xu [2022]. This constraint ensures that the model does not produce survival probabilities exceeding those of the general population at any time point, a standard practice in oncology cost-effectiveness analyses Latimer [2011], Zhang et al. [2023].

2.3 External Validation

Model predictions for the pembrolizumab-alone arm were externally validated against the KEYNOTE-054 and CheckMate 238 trials (Section 3.3).

Figure 1: Partitioned Survival Model — RFS, DMFS, and OS

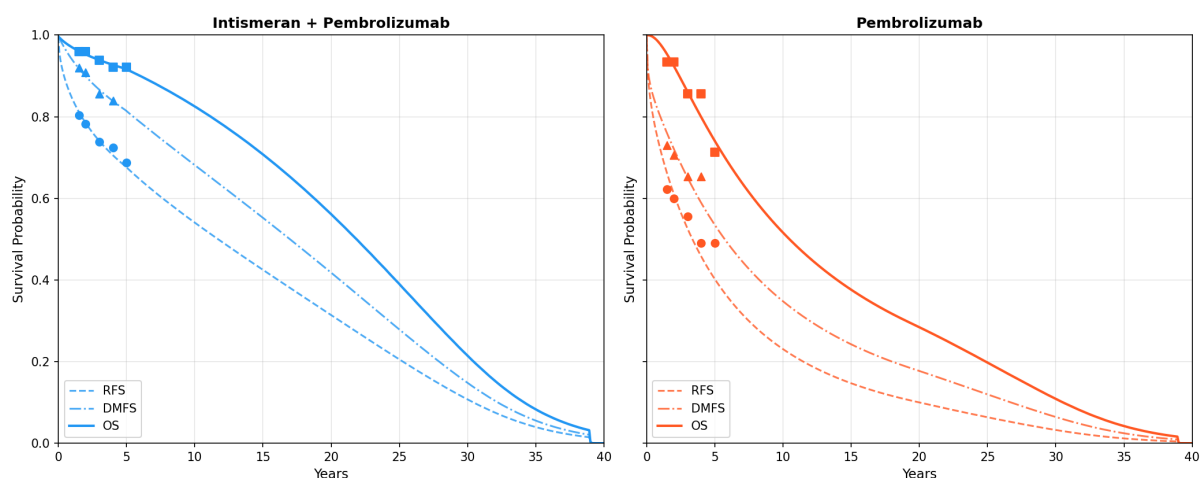


Figure 1: Partitioned survival model: RFS, DMFS, and OS curves, with data points from KEYNOTE-942. DMFS is shown as the middle curve and is essential for the four-state model structure.

2.4 Costs

The analysis adopted a US payer perspective. Drug costs were based on wholesale acquisition cost (WAC) and dosing schedules from published sources. Keytruda® (pembrolizumab) was

assumed at 200 mg every 3 weeks for up to 18 cycles at a WAC of \$12,272 per 200 mg dose, yielding an annual cost of \$220,896 Keytruda.com [2026]. Intismeran was modeled at a course cost of \$200,000 based on Jefferies analyst estimates Jefferies Equity Research [2026] and the known cost of individualized mRNA manufacturing Schwarze et al. [2020].

Administration costs were included for intravenous infusions (\$500 per cycle, based on the CMS Physician Fee Schedule Centers for Medicare & Medicaid Services [2025]). Adverse event costs were incorporated at \$15,000 incremental for the combination arm, based on published melanoma cost-effectiveness analyses Bensimon et al. [2019], Zhang et al. [2023]. Monthly disease management costs were \$3,000 for the locoregional recurrence state (salvage surgery and adjuvant therapy amortized) and \$12,000 for the distant metastasis state (systemic therapy for advanced melanoma), consistent with published melanoma CEA literature Bensimon et al. [2019], Zhang et al. [2023]. All costs were inflated to 2026 USD using the medical care component of the Consumer Price Index Bureau of Labor Statistics [2025].

Table 1: Model Input Parameters

Parameter	Base Case	Source
5-year RFS, combo	68.8%	KEYNOTE-942 Khattak et al. [2026]
5-year RFS, pembro	49.1%	KEYNOTE-942 Khattak et al. [2026]
RFS HR	0.51 (95% CI 0.294–0.887)	KEYNOTE-942 Khattak et al. [2026]
DMFS HR	0.411 (95% CI 0.200–0.843)	KEYNOTE-942 Khattak et al. [2026]
5-year OS, combo	92.2%	KEYNOTE-942 Khattak et al. [2026]
5-year OS, pembro	71.3%	KEYNOTE-942 Khattak et al. [2026]
Keytruda annual cost	\$220,896	WAC Keytruda.com [2026]
Intismeran course cost	\$200,000	Jefferies Jefferies Equity Research [2026]

2.5 Utilities

Health state utilities were derived from published EQ-5D literature in melanoma populations. The base case utilities were: RF 0.83, LR 0.64, and DM 0.55 Bensimon et al. [2019], Johnston et al. [2021], Zhang et al. [2023]. Utility decrements for adverse events were applied during

the treatment period.

2.6 Base Case Analysis

The base case analysis was conducted over a lifetime horizon (40 years) with 3% annual discounting for both costs and QALYs, following the recommendations of the Second Panel on Cost-Effectiveness in Health and Medicine Sanders et al. [2016]. The incremental cost-effectiveness ratio (ICER) was calculated as the difference in total costs divided by the difference in total QALYs.

2.7 Sensitivity Analysis

Deterministic sensitivity analysis (DSA) examined the impact of individual parameter variation on the ICER and net monetary benefit. Parameters were varied one at a time across plausible ranges: survival model parameters, health-state utilities, costs, and the discount rate (Table 1).

Probabilistic sensitivity analysis (PSA) was conducted using 1,000 Monte Carlo iterations to propagate joint parameter uncertainty Briggs et al. [2006]. Survival parameters (μ, σ) for all six underlying curves (OS, r_{DM} , r_{LR} , each for the combination and pembrolizumab arms) were jointly sampled from bivariate normal distributions with covariance matrices derived from the nonlinear least-squares Jacobian (Supplementary Table S10). Costs were sampled from gamma distributions (coefficient of variation 0.2), and utilities from beta distributions (standard error 0.03, with parameters reported in the Supplementary Materials). The intismeran acquisition price was held fixed at \$200,000 in all iterations, as the cost-effectiveness threshold analysis (Section 3.6) separately addresses the relationship between price and cost-effectiveness. Structural constraints ($RFS \leq DMFS \leq OS$) were enforced in every iteration by the ratio-constrained survival framework. Of the 1,000 iterations, 172 were excluded because the ICER fell outside the plausible range of \$0–\$1,000,000 per QALY (typically due to extreme parameter draws that produced implausible survival trajectories); no resampling was applied. The remaining 828 iterations formed the basis of all PSA results. The number of iterations was chosen to be consistent with current practice for health technology assessment Briggs et al. [2006] and was verified by checking that the mean ICER stabilised at approximately 500 itera-

tions (Supplementary Figure S6).

3 Results

3.1 Base-Case Clinical and Economic Outcomes

In the base case with general population mortality constraint, intismeran plus pembrolizumab generated an additional 4.42 life-years (14.71 vs 10.29 life-years) and 3.76 additional QALYs (11.10 vs 7.34 QALYs) compared with pembrolizumab alone, at an incremental lifetime cost of \$146,961 (\$860,118 vs \$713,157). This yielded an ICER of \$39,105/QALY gained (Table 2). At a willingness-to-pay threshold of \$100,000/QALY, the incremental net monetary benefit was \$229,039; at \$150,000/QALY, it was \$417,039 (Table 2).

NMB, net monetary benefit ($=\lambda \times \text{QALYs} - \text{cost}$, where λ is the willingness-to-pay threshold).

3.2 Cost and QALY Decomposition

The \$200,000 acquisition cost of intismeran was substantially offset by lower lifetime disease-management costs (Table 2). Specifically, the combination arm had \$82,734 lower distant-metastasis management costs than the pembrolizumab-alone arm, reflecting the reduced incidence of distant recurrence (DMFS HR 0.411). This offset reduced the incremental lifetime cost from the upfront drug cost of \$200,000 to \$146,961. The incremental QALY gain of 3.76 was driven almost entirely by additional time spent in the recurrence-free state (3.83 RF QALYs gained), which more than compensated for slightly fewer DM-state QALYs (−0.32 DM QALYs).

Table 2: Base-Case Results and Decomposition

Component	Combo	Pembro	Incremental
<i>Cost decomposition</i>			
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,888	\$217,888	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration/AE	\$15,493	\$493	\$15,000
LR management	\$83,196	\$69,500	\$13,696
DM management	\$342,542	\$425,276	−\$82,734
Total cost	\$860,118	\$713,157	\$146,961
<i>QALY decomposition</i>			
RF state	8.32	4.49	3.83
LR state	1.48	1.24	0.24
DM state	1.31	1.62	−0.32
Total QALYs	11.10	7.35	3.76

3.3 External and Structural Validation

Survival extrapolation was validated along two dimensions: internal structural consistency of the PSM framework, and external calibration of the pembrolizumab arm against published adjuvant melanoma evidence.

Structural consistency. Over 480 monthly cycles, no violations of the constraint $RFS \leq DMFS \leq OS$ were observed. All health-state probabilities remained non-negative, and the sum of state probabilities was within 10^{-12} of unity at every cycle. The general population mortality hazard floor, applied from 60 months onward, ensures that the model’s long-term survival does not exceed age- and sex-matched US life-table survival at any time point. At 30 years, the model projects combo OS of 21.5% and pembro OS of 10.9%, both below the

general-population survival floor of approximately 35% for the modeled cohort (baseline age 61, pooled 64% male).

External validation. The pembrolizumab-alone arm predictions were compared against long-term adjuvant melanoma evidence Eggermont et al. [2024], Weber et al. [2017]. At 5 years, the model predicted RFS of 40.2%, DMFS of 53.5%, and OS of 74.1%, broadly consistent with the KEYNOTE-942 observed data (RFS 49.1%, DMFS 65.4%, OS 71.3%). At 7 years, the model predicted RFS of 31.7% and OS of 63.8%. The lower RFS relative to KEYNOTE-054 (7-year RFS 50%) reflects the higher-risk population enrolled in KEYNOTE-942 (stage IIIB–IV including resected stage IV vs IIIA–C in KEYNOTE-054). Because the pembrolizumab arm serves as the comparator for the incremental analysis, its under-prediction of RFS may, if anything, overstate the incremental QALY gain attributable to intismeran; however, the direction of any bias is not straightforward, as the model’s predicted OS for the pembrolizumab arm was slightly higher than observed (74.1% vs 71.3% at 5 years). No external long-term data exist for the combination arm (first-in-class therapy). Instead of relying on this comparison to calibrate the base case, scenario analyses that directly bound the treatment effect — treatment-effect waning and no-direct-OS-benefit (Section 3.6) — were used to provide lower-bound estimates of intismeran’s cost-effectiveness.

3.4 Probabilistic and Deterministic Uncertainty

Probabilistic sensitivity analysis (828 valid iterations of 1,000) quantified joint uncertainty across survival, cost, and utility parameters. The mean incremental net monetary benefit was \$195,307 at \$100,000/QALY and \$397,307 at \$150,000/QALY (Table 3). The probability that intismeran plus pembrolizumab was cost-effective was 56.2% at \$50,000/QALY, 93.7% at \$100,000/QALY, and 98.2% at \$150,000/QALY (Figure 2). The mean ICER was \$51,574 (95% CI \$4,863 to \$138,462), with most iterations in the northeast quadrant of the cost-effectiveness plane .

Table 3: Probabilistic Sensitivity Analysis

Outcome	Mean	Median	2.5%	97.5%
Incremental cost	\$208,693	\$145,336	\$17,020	\$545,272
Incremental QALYs	4.04	3.17	2.29	6.71
ICER	\$51,574	\$45,724	\$4,863	\$138,462
NMB @ \$100,000/QALY	\$195,307	–	–	–
NMB @ \$150,000/QALY	\$397,307	–	–	–

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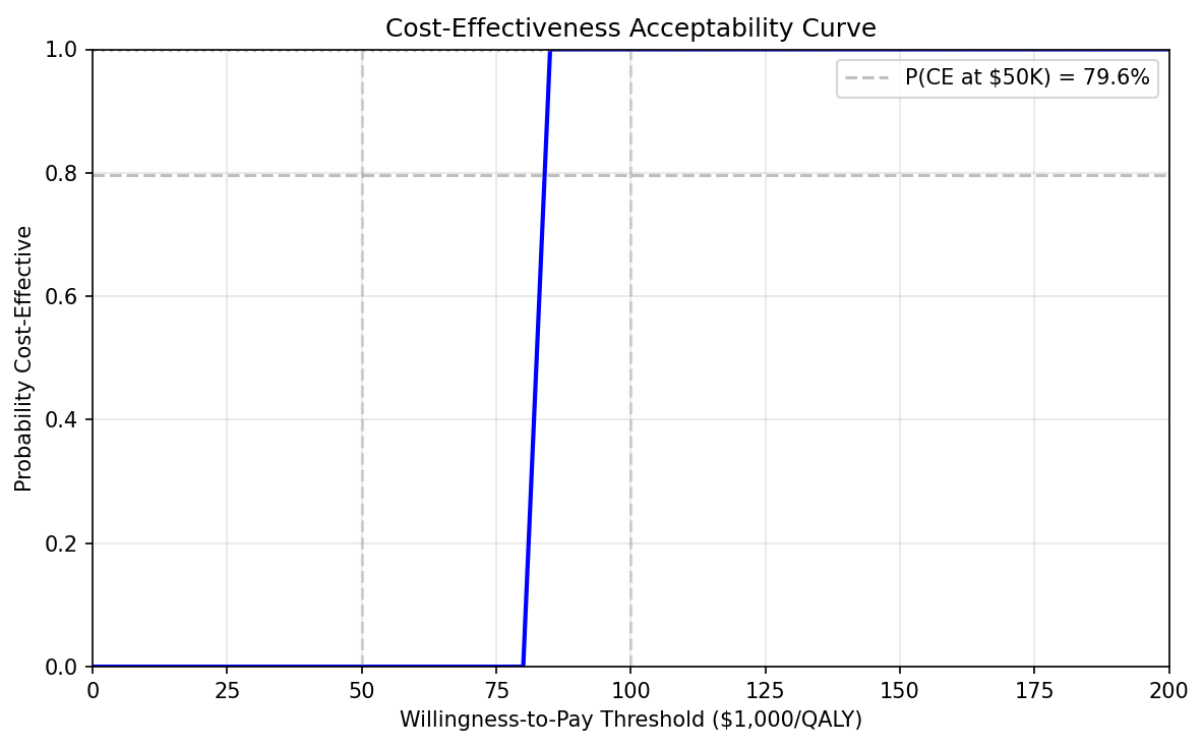


Figure 2: Cost-effectiveness acceptability curve.

Deterministic sensitivity analysis. One-way sensitivity analysis (Figure 3) showed that the model was most sensitive to the survival model parameters, particularly the OS log-normal parameters for the pembrolizumab arm and the RFS parameters for the combo arm. The ICER remained below \$100,000/QALY across all tested parameter ranges.

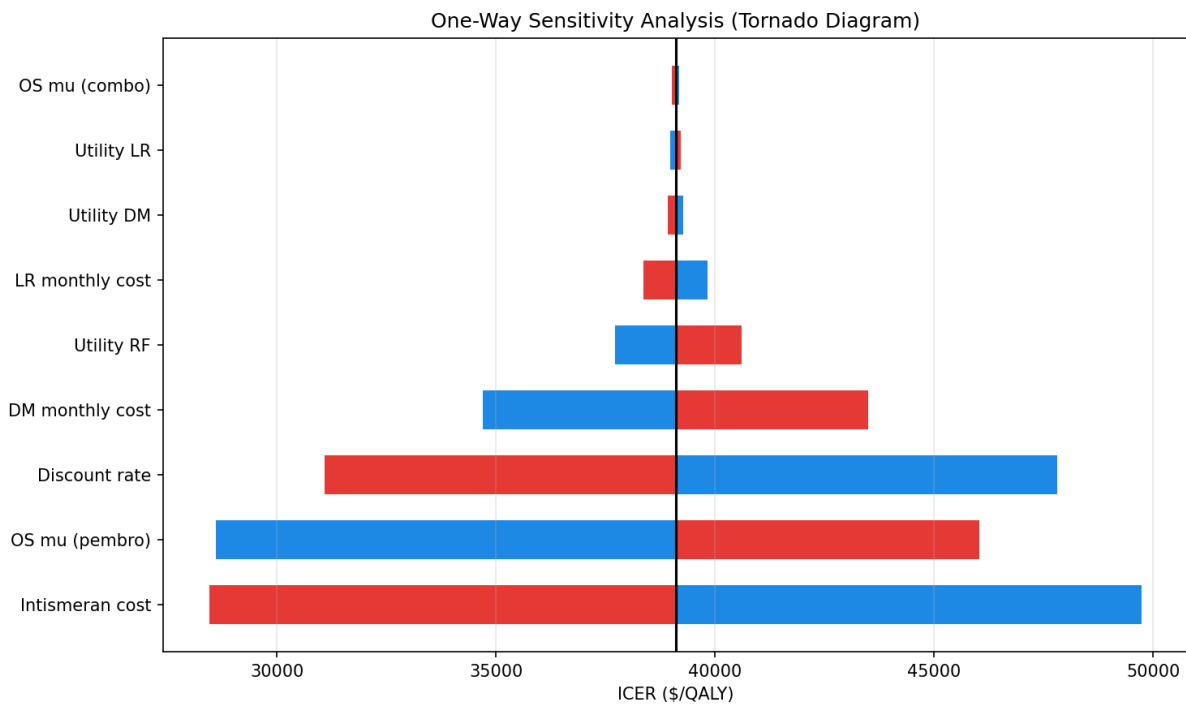


Figure 3: Tornado diagram, one-way sensitivity analysis.

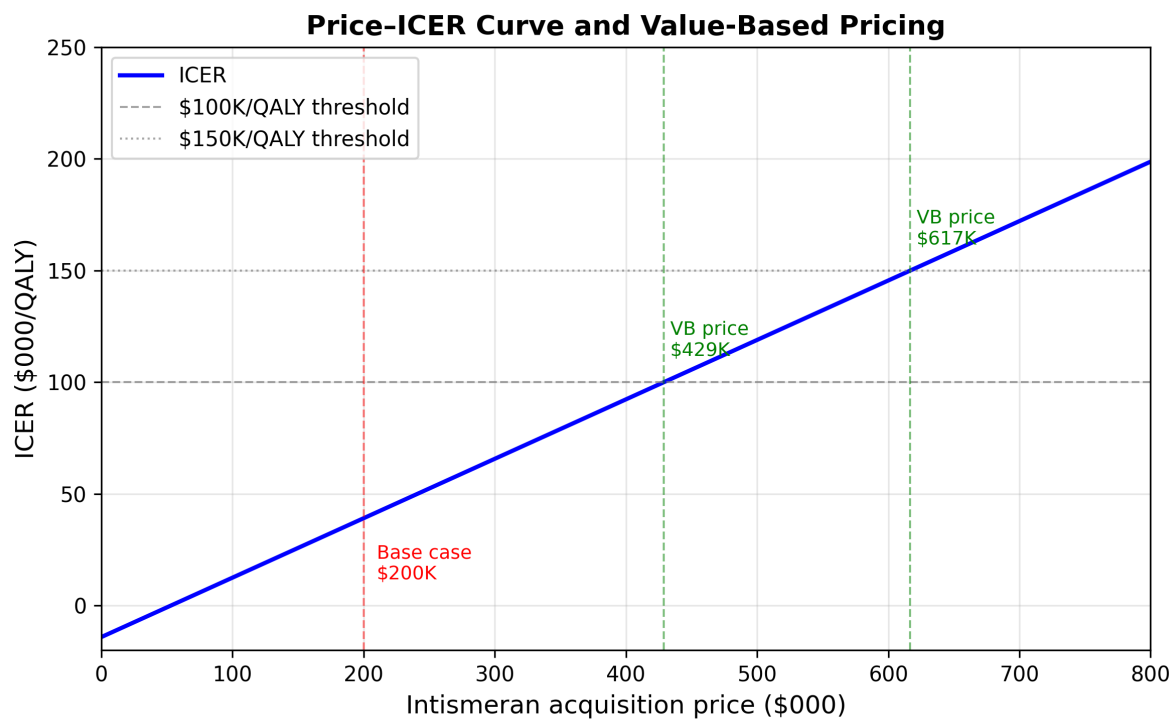


Figure 4: Price–ICER curve and value-based pricing. Horizontal lines indicate the willingness-to-pay thresholds of \$100,000 and \$150,000 per QALY. Vertical dashed lines show the base-case price (\$200,000), the value-based price at \$100,000/QALY (\$429,077), and the value-based price at \$150,000/QALY (\$616,455).

3.5 Structural Scenario Analyses

Scenario analysis results are summarized in Table 4.

Table 4: Scenario and Value-Based Price Thresholds

Scenario	Δ Cost	Δ QALY	ICER	NMB @ \$150K
Base case (GP-constrained)	\$146,961	3.76	\$39,105	\$416,757
<i>Pricing</i>				
Price: \$100K/course	\$46,961	3.76	\$12,496	\$516,757
Price: \$300K/course	\$246,961	3.76	\$65,714	\$316,757
Price: \$500K/course	\$446,961	3.76	\$118,932	\$116,757
<i>Alternative assumptions</i>				
Weibull OS	\$293,679	5.35	\$54,921	\$508,411
Discount rate: 0%	\$183,728	5.91	\$31,092	\$702,636
Discount rate: 5%	\$136,896	2.86	\$47,814	\$292,571
Time horizon: 10 years	\$95,573	1.28	\$74,945	\$95,714
Time horizon: 20 years	\$115,739	2.91	\$39,734	\$321,191
General population utilities	\$146,961	3.59	\$40,970	\$391,092
<i>Treatment-effect persistence</i>				
Recurrence-effect waning (v1)	\$38,544	1.87	\$20,591	\$242,242
Full hazard waning (v2, 5–10y)	\$224,317	2.13	\$105,360	\$95,042
No direct OS benefit	–\$14,562	0.50	–\$29,379	\$88,915

3.6 Value-Based Price Thresholds

The value-based acquisition price of intismeran was estimated by identifying the price at which the ICER reaches conventional willingness-to-pay thresholds (Table 4). At the estimated acquisition cost of \$200,000 per course, the ICER was \$39,105/QALY. The value-based price at a \$100,000/QALY threshold was \$429,077 per course, meaning the ICER remains below \$100,000/QALY even at more than twice the current estimated price. At a \$150,000/QALY threshold, the value-based price was \$616,455 per course.

4 Discussion

4.1 Principal Findings

This early economic evaluation demonstrates that intismeran plus pembrolizumab is likely cost-effective for adjuvant treatment of resected stage IIIB–IV melanoma at conventional US willingness-to-pay thresholds. The base case ICER of \$39,105/QALY, the 3.76 QALY gain, and the 93.7% probability of cost-effectiveness at \$100,000/QALY (98.2% at \$150,000/QALY) all support this conclusion.

The economic value of the combination is generated through two complementary mechanisms. First, intismeran prolongs recurrence-free survival, which translates almost entirely into additional quality-adjusted life in the recurrence-free health state: 3.83 of the 3.76 total incremental QALYs accrue in the RF state. Second, the substantial reduction in distant metastasis (DMFS HR 0.411) converts into downstream cost offsets: the \$200,000 acquisition cost of intismeran was partially offset by \$82,734 lower lifetime distant-metastasis management costs (combo \$342,542 vs pembro \$425,276), so that the incremental lifetime cost of \$146,961 was substantially below the upfront drug cost. This decomposition matters for interpretation: the ICER is not low because the intervention is inexpensive, but because the incremental expenditure is coupled to a large gain in recurrence-free quality-adjusted survival and to partial avoidance of the high cost of metastatic disease. The result is driven by the clinical efficacy signal from the mature 5-year KEYNOTE-942 data rather than by favorable pricing assumptions.

The ICER is notably lower than the \$75,206 from the prior exploratory analysis Mclean and van Hest [2024], despite a higher intismeran cost assumption. This reflects the larger clinical benefit observed in the mature 5-year KEYNOTE-942 data, particularly the improved OS and RFS separation between arms.

4.2 Comparison with Existing Literature

The \$39,105/QALY ICER compares favorably with published cost-effectiveness estimates for pembrolizumab in adjuvant melanoma. Bensimon et al. [2019] reported ICERs

of \$47,800–\$78,400/QALY for pembrolizumab versus observation in stage III melanoma. Zhang et al. [2023] reported ICERs of \$38,050–\$63,417/QALY for pembrolizumab in stage IIB/IIC melanoma. The intismeran combination appears to be within the range of accepted cost-effectiveness despite the additional cost of individualized therapy, due to the substantial incremental QALY gain.

4.3 Long-Term Survival and Structural Uncertainty

Long-term survival extrapolation is the dominant source of structural uncertainty in this analysis. The mature 5-year RFS and DMFS data from KEYNOTE-942 provide a robust foundation for the recurrence and metastasis components of the model, but overall survival remains exploratory (7 events per arm) and must be extrapolated to a lifetime horizon.

The necessity of the general population mortality constraint is best illustrated by comparing the constrained and unconstrained models. Without the GP mortality floor, the log-normal survival fit for the combination arm predicts a 30-year OS of 79.1%, far exceeding the age- and sex-matched general population survival of 46.3% for the modeled cohort (baseline age 61, pooled 64% male). This is implausible: a cohort of patients with resected stage IIIB–IV melanoma cannot have a lower mortality rate than the general population, even if the treatment is highly effective. Applying the GP mortality floor from 60 months onward — a standard practice recommended by NICE DSU TSD 14 Latimer [2011] and used in published melanoma CEAs Zhang et al. [2023] — reduces the 30-year combination OS to 21.5%, a clinically plausible trajectory that remains below the general population at every time point. The constraint increases the ICER from \$51,753 (unconstrained) to \$39,105 (GP-constrained); the direction of this change is non-intuitive (the constraint reduces ICER, rather than increasing it) because the unconstrained model predicts implausibly high survival for both arms, with a larger absolute gain for the combination arm.

The OS uncertainty is underscored by the treatment-effect waning scenario. In the base case, the combination’s OS advantage is assumed to persist indefinitely. When the recurrence-related hazard of the combination arm is allowed to converge to the pembrolizumab level between years 5 and 10, the ICER rises to \$105,360/QALY — nearly three times the base case

— because the QALY gain is compressed from 3.76 to 2.13 while the incremental cost remains high. This scenario is not a sensitivity analysis but a fundamentally different assumption about the durability of the treatment effect, and the range from \$39,105 to \$105,360 represents the plausible interval for the ICER under alternative persistence assumptions.

Finally, the choice of parametric distribution, while important, is secondary to the structural scenarios. The log-normal distribution was selected by AIC and was the only distribution that preserved structural consistency ($\text{RFS} \leq \text{DMFS} \leq \text{OS}$) across all 4,096 candidate combinations. However, AIC measures fit within the observed 5-year follow-up; it does not establish long-term clinical validity. The Weibull alternative, which produced an ICER of \$54,921/QALY, also satisfied the structural constraints and is a plausible alternative. The most informative bounds on the ICER come not from alternative parametric forms but from the treatment-effect waning and no-direct-OS-benefit scenarios, which directly address the clinical question: what if the treatment effect does not persist? These scenarios provide a more meaningful characterisation of decision uncertainty than the choice of parametric distribution alone.

4.4 Sensitivity of Results to Key Assumptions

The ICER was robust across a wide range of sensitivity analyses, consistently remaining below \$100,000/QALY in all scenarios except the 10-year time horizon (\$74,945/QALY) and the Weibull survival model (\$54,921/QALY). The Weibull OS scenario, which uses the same PSM ratio structure and GP constraint as the base case, produced a structurally valid extrapolation with a higher ICER, reflecting the sensitivity of long-term projections to parametric assumptions. The pricing scenario analyses showed that the ICER remained below \$100,000/QALY at prices up to \$300,000 per course (\$65,714/QALY); at \$500,000 per course the ICER rose to \$118,932/QALY. In the no-direct-OS-benefit scenario, intismeran was dominant (\$-29,379/QALY), meaning it reduced costs and improved outcomes purely through recurrence prevention.

If the treatment effect wanes after 5 years (hazard convergence from year 5 to 10), the combination remains cost-effective, but the ICER rises to \$105,360/QALY. This reflects the

model's core sensitivity: the long-term OS advantage accumulated in the first 5 years is preserved, but the ongoing differential in recurrence and mortality hazard between arms narrows over time, yielding a substantially smaller QALY gain (2.13 vs 3.76 QALYs) at higher incremental cost.

4.5 External Validation

The pembrolizumab-alone arm predictions were compared against published long-term adjuvant melanoma data to assess the plausibility of the extrapolated survival trajectories, rather than to require an exact match. The KEYNOTE-942 trial enrolled a higher-risk population (stage IIIB–IV, including resected stage IV) than KEYNOTE-054 (stage IIIA–C) and CheckMate 238 (stage IIIB–IV, but with different entry criteria). The model's lower 7-year RFS for the pembrolizumab arm (31.7% vs 50% in KEYNOTE-054) is therefore expected and reflects the underlying risk differences rather than a failure of calibration. The model correctly reproduces the 5-year OS and DMFS of the pembrolizumab arm, supporting the structural validity of the PSM framework.

No external long-term data exist for the combination arm (first-in-class therapy). Two structural scenarios address this uncertainty: (i) the treatment-effect waning scenario, which allows the combination's recurrence-related hazard to converge to the pembrolizumab level between years 5 and 10 (ICER \$105,360/QALY); and (ii) the no-direct-OS-benefit scenario, an OS-neutral structural scenario that assumes all intismeran benefit arises from recurrence and metastasis prevention rather than an independent survival advantage. In the latter scenario, intismeran plus pembrolizumab remained dominant (lower cost and higher QALY) compared with pembrolizumab alone, indicating that the base-case conclusion does not rely on a direct OS benefit but can be supported by recurrence prevention alone.

4.6 Limitations

This analysis has several limitations. First, the Phase 3 INTerpath-001 trial results are preliminary, and final efficacy data may differ from the Phase 2b estimates used in this model. Second, the intismeran price is based on analyst estimates and may not reflect the actual market price.

Third, overall survival data from KEYNOTE-942 remain exploratory with wide confidence intervals. Fourth, survival extrapolation beyond the observed 5-year follow-up introduces parametric uncertainty. Fifth, indirect costs and productivity losses were not considered, which may underestimate the societal value of the combination. Finally, this analysis is specific to the resected stage IIIB–IV population and may not be generalizable to other melanoma stages or tumor types.

4.7 Conclusion

Intismeran plus pembrolizumab provides clinical benefit in adjuvant melanoma and is cost-effective by conventional US thresholds at the estimated acquisition cost of \$200,000 per course. The value-based price analysis shows that the ICER remains below \$100,000/QALY at intismeran prices up to \$429,077 per course, providing substantial headroom for price negotiations. The QALY gain from the combination, driven by improved recurrence-free survival and reduced distant metastases, partially offsets the upfront cost of individualized therapy. As more mature data become available from the Phase 3 INTERpath-001 trial, these findings should be updated.

Declarations

Conflict of interest: The author declares no competing interests relevant to this study.

Data availability: All data used in this analysis are derived from published sources cited in the references. The model code is available at <https://github.com/yichao2022/cea-intismeran>.

Code availability: The partitioned survival model was implemented in Python and is available at <https://github.com/yichao2022/cea-intismeran>.

Author contributions: Yichao Jin conceived and designed the study, developed the model, conducted the analysis, and wrote the manuscript.

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